TECHNICAL NOTE: The MVS Framework
Underwriting External Credit via Digitized ROSCA Behaviour

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Subject: Operationalizing ROSCA Behaviour for Initial Credit Scoring and PD Calibration

1 The narrative: the information problem and why ROSCAs matter

The central constraint to extending credit in financially thin environments is information asymmetry. When bureau histories are incomplete or absent, lenders cannot directly observe repayment type. Rotating Savings and Credit Associations (ROSCAs)—tontines, osusus—matter because they partially solve this information problem through repeated interaction, peer observability, and embedded discipline. The theoretical foundation is clear: ROSCAs can be understood as repeated-game institutions with endogenous enforcement **BesleyCoateLoury1993<empty citation>**. Empirically, ROSCAs often function as commitment devices that help members maintain disciplined periodic payments **Gugerty2007<empty citation>**. Governance quality, sanctions, and social connectedness shape enforcement and sustainability over time **McNabbLeMayBoucherBonan2019<empty citation>**. In bidding ROSCAs, the bid discount behaves like an implicit interest rate and reveals liquidity stress and shocks **Klonner2003**; **CalomirisRajaraman1998<empty citation>**. A key bridge to formal underwriting is the insight that *publicly observable characteristics* inside these informal institutions can predict creditworthiness. **SchreinerNagarajan2000<empty citation>** provide direct evidence from Gambian ASCRAs/ROSCAs that cheap, observable reliability markers contain predictive signal for credit screening. This is precisely the operational premise of MVS: turn socially embedded observables into an interpretable underwriting signal at launch, then calibrate it into a formal probability of default once outcomes accumulate.

2 Literature-grounded mechanisms: what we are measuring

The score is intentionally mechanism-based rather than purely correlational. We map observed ROSCA behaviour to economic channels identified in the literature. Payment discipline and persistence capture time-consistent commitment **Gugerty2007<empty citation>**. Allocation order and post-payout slippage encode incentive compatibility and moral hazard after receiving the pot **BesleyCoateLoury1993<empty citation>**. Governance and enforcement strength reflect heterogeneity in rules, sanctions, and connectedness **McNabbLeMayBoucherBonan2019<**
In bidding ROSCAs, discount level and volatility reveal liquidity stress in the upper tail **Klonner2003**; **CalomirisRajaraman1998<empty**
Finally, social capital proxies (repeat interaction, centrality, endorsements) raise reputational stakes and strengthen informal enforcement **SchreinerNagarajan2000<empty citation>**.

3 Target, label, and decision-time discipline (anti-leakage)

The long-run objective is to estimate PD@3M: the probability that a loan reaches DPD60 within 90 days after disbursement. We define the calibration label:

$$\text{Def3M}_i = \mathbb{I}\{\text{DPD60 occurs on or before day 90 post-disbursement}\}. \tag{1}$$

All predictors must be computed using only information available at the underwriting decision timestamp T_i. This "decision-time discipline" prevents leakage and ensures deployability.

Because payout timing matters, define:

$$\text{AllocByDec}_i = \mathbb{I}\{\text{AllocDate}_i \leq T_i\}. \tag{2}$$

Post-payout features are valid only when $\text{AllocByDec}_i = 1$.

4 Data dictionary (minimum viable schema)

To keep the system auditable and partner-ready, we define a minimum viable schema. Variable names are intentionally short for operational simplicity. All timestamps are required to respect the decision-time cutoff.

Field	Level	Type	Definition
mid	member	string	Member identifier
gid	group	string	ROSCA/group identifier

Field	Level	Type	Definition
cid	cycle	string	Cycle identifier
mdate	meeting	date	Monthly meeting date
t_dec	decision	datetime	Underwriting decision timestamp (feature cutoff)
due	meeting	number	Amount due
paid	meeting	number	Amount paid
dlate	meeting	int	Days late (0 if on time)
ont	meeting	{0,1}	On-time flag: 1 if dlate$=0$
N	cycle	int	Group size
aord	member/cycle	int	Allocation order position (1..N)
adate	member	date	Allocation date (pot receipt)
rtype	group	enum	ROSCA type: RANDOM / BIDDING / COMMITTEE
rules	group	{0,1}	Documented rules exist and communicated
san6	member/group	int	Count of sanctions in last 6 meetings
santype	meeting	enum	LATE_FEE / ABSENCE_FINE / WARNING / SUSPENSION / OTHER
sanamt	meeting	number	Monetary amount of sanction
sure_n	member	int	Number of sureties (0/1/2+)
sure_type	member	enum	NONE / INDIVIDUAL / POOLED
sure_str	member	enum	NONE / WEAK / STRONG
bid_flag	attempt	{0,1}	Bid attempt made
disc	attempt	number	Bidding discount (%) recorded per attempt
rep	member	{0,1}	Repeat participation in same group across cycles
cent	member	{0,1}	Centrality above group median
end_f	member	{0,1}	Foreman endorsement (clean)
end_s	member	{0,1}	Senior endorsement (clean)
loanid	loan	string	Loan identifier
t_disb	loan	datetime	Disbursement timestamp
ten	loan	int	Tenor (days)
freq	loan	enum	WEEKLY / BIWEEKLY / MONTHLY
dpd60	loan	{0,1}	DPD60 flag
def3m	loan	{0,1}	Def3M label (defined above)

5 From mechanisms to measurement (narrative $\rightarrow$ guided math)

The design objective of MVS is to remain interpretable while reflecting the non-linear nature of behavioural risk. Early history is disproportionately informative relative to incremental history; tail volatility and clustered slippage matter more than small average differences. We therefore use smooth, monotone non-linear maps—sigmoid, exponential decay, and log transforms—so that the score saturates where it should and becomes sharply sensitive where risk concentrates.

5.1 Eligible window and time decay

Let K_i be the number of eligible monthly meetings prior to T_i (standard track: $K_i \geq 6$). Index meetings $m = 1, \dots, K_i$ (oldest $\rightarrow$ newest). Define time-decay weights:

$$w_{im} = \lambda^{K_i - m}, \qquad 0 < \lambda < 1, \tag{3}$$

with a conservative initial setting $\lambda = 0.80$ for monthly cadence (sensitivity-tested during calibration).

5.2 Time-decayed behavioural primitives

From meeting-level primitives, compute time-decayed on-time rate and average lateness:

$$W_i = \sum_{m=1}^{K_i} w_{im}. \tag{4}$$

$$\mathrm{OTR}_i = \frac{1}{W_i} \sum_{m=1}^{K_i} w_{im}\, \mathrm{ont}_{im}, \qquad \mathrm{AL}_i = \frac{1}{W_i} \sum_{m=1}^{K_i} w_{im}\ \max(0, \mathrm{dlate}_{im}). \tag{5}$$

Let LS_i be the maximum consecutive late streak in the eligible window.

We use:

$$\mathrm{sigmoid}(x) = \frac{1}{1 + e^{-x}}, \qquad \mathrm{log1p}(x) = \ln(1 + x). \tag{6}$$

5.3 Pillar 1: Payment discipline (PDIS)

Map primitives into smooth monotone components:

$$S_i^{OTR} = 20 \cdot \mathsf{sigmoid}(k_{OTR}(\mathsf{OTR}_i - c_{OTR})), \tag{7}$$
$$S_i^{AL} = 6 \cdot \mathsf{exp}(-a_{AL} \cdot \mathsf{log1p}(\mathsf{AL}_i)), \tag{8}$$
$$S_i^{LS} = 9 \cdot \mathsf{exp}(-a_{LS} \cdot \mathsf{LS}_i). \tag{9}$$

A small recovery term S_i^{RC} ("rapid cure") can be included when late/missed payments are cured within 7 days.

Normalize to a 35-point scale:

$$S_i^{PDIS} = \frac{35}{37} \left(S_i^{OTR} + S_i^{AL} + S_i^{LS} + S_i^{RC} \right). \tag{10}$$

5.4 Pillar 2: Order and post-payout incentives (ORDR)

Define allocation bucket $\mathsf{AB}_i \in \{\mathsf{EARLY}, \mathsf{MID}, \mathsf{LATE}\}$ using terciles of aord/N. Assign base multipliers:

$$b(\mathsf{LATE}) = 1.0, \quad b(\mathsf{MID}) = 0.6, \quad b(\mathsf{EARLY}) = 0.3. \tag{11}$$

Define post-payout slippage Slip_i as an indicator equal to 1 if a new late streak of length ≥ 2 begins after allocation and before T_i (valid only when $\mathsf{AllocByDec}_i = 1$). Then:

$$S_i^{ORDR} = 15 \cdot b(\mathsf{AB}_i) \cdot (1 - \pi_{Slip} \cdot \mathsf{Slip}_i). \tag{12}$$

5.5 Pillar 3: Governance and enforcement (GOV)

Map rules and sanctions via monotone functions:

$$S_i^{Rules} = 5 \cdot \mathsf{sigmoid}(k_R(\mathsf{rules}_i - 0.5)), \tag{13}$$
$$S_i^{San} = 6 \cdot \mathsf{exp}(-a_Z \cdot \mathsf{san6}_i). \tag{14}$$

Let $S_i^{Sure} \in \{0, 3, 6\}$ map surety strength NONE/WEAK/STRONG. Normalize:

$$S_i^{GOV} = \frac{20}{17} \left(S_i^{Rules} + S_i^{San} + S_i^{Sure} \right). \tag{15}$$

5.6 Pillar 4: Liquidity stress in bidding ROSCAs (LIQ)

For bidding ROSCAs, let A_i be bid attempts in the eligible window. Define attempt volatility via interquartile range:

$$\text{IQR}_i = Q_{75}(\text{disc}_{ia}) - Q_{25}(\text{disc}_{ia}), \quad a \in A_i. \tag{16}$$

Let $q_i \in [0,1]$ be the member's discount quantile rank within the group attempt distribution. Define:

$$S_i^{Lvl} = 6 \cdot (1 - \text{sigmoid}(k_q(q_i - q_0))), \tag{17}$$

$$S_i^{Vol} = 6 \cdot \exp(-a_v \cdot \text{log1p}(\text{IQR}_i / v_{ref})), \tag{18}$$

$$S_i^{LIQ} = \frac{15}{12}\left(S_i^{Lvl} + S_i^{Vol}\right). \tag{19}$$

5.7 Pillar 5: Social capital (SOC)

Use a bounded linear combination of cheap observables:

$$\text{socraw}_i = 4\,\text{rep}_i + 3\,\text{cent}_i + 2\,\text{end\_f}_i + 2\,\text{end\_s}_i, \qquad S_i^{SOC} = \frac{15}{11}\,\text{socraw}_i. \tag{20}$$

5.8 Final score and guardrails

Compute the raw total:

$$\text{MVS}^{*raw}_i = S_i^{PDIS} + S_i^{ORDR} + S_i^{GOV} + S_i^{LIQ} + S_i^{SOC}. \tag{21}$$

Guardrails are applied last to prevent tail-risk dilution and structural blind spots. Decision bands (pre-calibration): Green (MVS$^* \geq 75$), Amber ($60 \leq$ MVS$^* \leq 74$), Red (MVS$^* < 60$).

6 Strengths, limits, and operational use

MVS is interpretable: each pillar corresponds to a literature-backed mechanism, enabling transparent reason codes for approvals/declines. It also addresses cold start by enabling underwriting without formal bureau history using repeated behavioural data.

At launch, MVS is not a calibrated PD; it is a decisioning and ranking tool. Accept-only bias is a key risk if lending occurs only to high-score applicants. These limitations are addressed through a governed Phase 1 pilot.

7 The calibration imperative (phases and why Phase 1 must be governed)

Stable PD calibration requires outcome data and sufficient default events; event counts, not only loan counts, determine model stability **RileyEtAl2020**; **VittinghoffMcCulloch2006<empty citation>**. Overly complex models too early risk overfitting. The appropriate approach is phased:

Phase 0 (data readiness): lock taxonomy, schema, timestamps, QA rules, and ensure MVS can be computed deterministically.

Phase 1 (go-live with controlled exploration): deploy MVS with guardrails and reason codes; approve conservatively while allocating a bounded learning budget to generate outcomes beyond the acceptance frontier.

Phase 2 (first calibration): fit parsimonious logistic and/or survival models mapping MVS (and a small set of primitives) to Def3M; validate out-of-time and by group clusters (gid).

Phase 3 (scale and sophistication): as event volume grows, move to monotone GAMs and a monotone GBM challenger; refresh calibration regularly and monitor drift. Probability calibration should be monitored explicitly (see **SklearnCalib** for standard calibration diagnostics).

8 Phase 1 pilot governance: numeric limits a partner can sign

To bound downside while enabling learnable outcomes, the first 500 loans should run under explicit numeric governance constraints. The Core sleeve is at least 85% of approvals (Green and high-Amber). The Exploration sleeve is capped at 15% (low-Amber and a small, selected subset of high-potential Red profiles).

Let E_0 be the standard initial exposure cap agreed with the partner. In Phase 1, per-borrower maximum exposure is capped at E_0 in Core and $0.35E_0$ in Exploration. Products should generate delinquency signals within 90 days: tenor 90–120 days with weekly or biweekly installments.

Hard stop-loss triggers are applied mechanically. Pause new Exploration approvals if Exploration DPD30 exceeds 12% over the most recent 60 disbursement days, or if Exploration DPD60 exceeds 6%. Pause all new approvals if portfolio-level DPD60 exceeds 4% over the most recent 60 disbursement days. Outcomes are monitored by weekly vintages, and each decision stores pillar-level reason codes and guardrail flags.

9 Strategic positioning (partner discussion)

This approach is not merely a formula; it is an engineered learning loop. We begin with a mechanism-based, auditable score grounded in ROSCA theory and evidence **BesleyCoateLoury1993**; **Gugerty2007**; **McNabbLeMayBoucherBonan2019**; **Klonner2003**; **CalomirisRajaraman1998**; **SchreinerNagarajan2000<empty citation>**. We go live conservatively with guardrails and reason codes, run a governed pilot with bounded exploration, and then calibrate PD@3M with parsimonious models before scaling, consistent with event-driven methodology **RileyEtAl2020**; **VittinghoffMcCulloch2006<empty citation>**.

References